Geometry
Unit 9
Lesson 2 Homework

Name___Answer Key_______
Date____________________
Period__________________

The table includes the populations of a few cities in Florida.

City	Population in 2014	Population in 2019
Destin	12,840	13,702
Panama City	36,405	36,640
Pensacola	52,505	52,642

1. Pensacola is spread over an area of 41.12 square miles. What was the population density in 2014?

$\frac{52505}{41.12} \approx 1277 \frac{\text{people}}{\text{sq. mile}}$

2. What was the population density in 2019?

$\frac{52642}{41.12} \approx 1280 \frac{\text{people}}{\text{sq. mile}}$

3. How did the population density change over the course of 5 years?

It increased by about 3 people per square mile.

4. Destin is spread over an area of 7.66 square miles and Panama City covers 41.27 square miles. What was the population density of Destin in 2019?

$\frac{13702}{7.66} \approx 1789 \frac{\text{people}}{\text{sq. mile}}$

5. What was the population density of Panama City in 2019?

$\frac{36640}{41.27} \approx 888 \frac{\text{people}}{\text{sq. mile}}$

6. How does the population density of Destin compare with the population density of Panama City?

Destin has about 901 more people per square mile.

7. Manatee Springs is one natural spring that provides warm water for manatees to survive. The spring run that allows water to flow from the spring into the Suwannee River is $\frac{1}{4}$ mile long. There are about 20 manatees that can be seen regularly here. If the spring run is $\frac{1}{4}$ mile wide, what is the population density of the manatees?

$\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$

$\frac{20}{\frac{1}{16}} = 320$ manatees/sq. mile

8. Miami is a popular tourist spot due to its beaches and warm weather. It spreads across an area of 55.25 square miles. If 24.2 million people visit the area, what is the density of visitors?

$\frac{24.2 \text{ million}}{55.25 \text{ sq. miles}} \approx 0.438$ million visitors/sq. mile

9. If only $\frac{1}{3}$ of those visitors are in Miami during the winter months, what is the population density of visitors?

$\frac{8.067 \text{ million}}{55.25 \text{ sq. miles}} = 0.146$ million/sq. mile

10. If $\frac{3}{4}$ of the visitors are in Miami during the summer months, what is the population density of visitors?

$\frac{18.15 \text{ million}}{55.25 \text{ sq. miles}} \approx 0.329$ million/sq. mile